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Question 1

Correct

Mark 4.00 out of 4.00

Obstructive techniques

For each of the following examples, identify the type of obstructive techniques that may hamper the progress of information sharing;

Do you take a proper break from work every six months and are you able to relax?	Polythematic questions ✔
These voices, which you are describing, are not from your head. Am I right?	Suggestive questions ✔
When the patient stated, 'Over the last four weeks I have had trouble with my sleep', the doctor replied 'We are here to talk about your mood.'	Disapproval ✔
You wanted to see me as nothing had gone well for you. But you just mentioned that you have got a new job and entered into a new relationship.	Setting traps ✔

Your answer is correct.

Explanation:

Compound or **polythematic questioning** is adding two or more questions in a single statement. This interview technique confuses the patient and will lead to either a vague response or non-response.

In **suggestive questions**, the answers are contained in the question. It misleads both the patient and the doctor. The patient is left with little choice.

Disapproval is expressing unhappiness with a topic that the patient wants to discuss; may lead to withdrawal and not revealing the important problem faced by the patient.

Setting traps is tricking the patient using his words. Often seen as doctor's attempt to negate patient's problems.

The correct answer is: Do you take a proper break from work every six months and are you able to relax? → Polythematic questions, These voices, which you are describing, are not from your head. Am I right? → Suggestive questions, When the patient stated, 'Over the last four weeks I have had trouble with my sleep', the doctor replied 'We are here to talk about your mood.' → Disapproval, You wanted to see me as nothing had gone well for you. But you just mentioned that you have got a new job and entered into a new relationship. → Setting traps

Question 2

Correct

Mark 3.00 out of 3.00

Electrolyte abnormality
What electrolyte abnormality could be expected in each of the following situations?

1. Rani, a 52-year-old woman under the oncologists for breast cancer complains of being excessively thirsty. Hypercalcaemia ✔

2. Jay, a 22-year-old man, was brought into A&E in a coma. His friend Josh said that they had been at a rave where he thinks Jay may have taken something. He describes Jay as being very chatty and overfamiliar. He said that because Jay had a lot of alcohol before, his friends encouraged him to drink lots of water to sober up. Hyponatraemia ✔

3. Lena, a 19-year-old woman, is under her GP for low mood. In the past she suffered from bulimia and struggles with her weight. Her boyfriend broke up with her a few days ago. Her friend Jess brings her to A&E as Jess noticed Lena was a little confused. She also has a tremor and some numbness and is dizzy. Metabolic alkalosis ✔

Explanation:

1) Hypercalcaemia - polydipsia is a common symptom of hypercalcaemia. Hypercalcaemia is relatively common in cancer (20-30% of cancer patients develop hypercalcaemia). The most common cancers associated with hypercalcaemia are breast and lung cancer, and multiple myeloma.

2) Hyponatraemia - Jay has likely taken MDMA which results in chatty and overfamiliar behaviour, and excessive thirst. He has developed hyponatreamia secondary to over-drinking water due to this excessive thirst. MDMA also causes syndrome of inappropriate antidiuretic hormone (SIADH) which contributes to hyponatremia.

3) Metabolic alkalosis - metabolic alkalosis is associated with excessive vomiting. In this case, it is likely that Lena has started to binge and purge following the breakdown of her relationship putting her at risk of developing metabolic alkalosis.

Question 3

Correct

Mark 3.00 out of 3.00

Movement disorders
Choose the movement disorder that matches the description.

1. A man who kicks his wife once as he falls asleep. Sleep myoclonus/hypnic jerk ✔

2. A man on psychotropic medication for a long time who sleeps badly, wakes with a stiff back in the morning and takes at least 10 minutes to get out of bed. Parkinsonism ✔

3. A man who experiences slow writhing movements in peripheral limbs. Athetosis ✔

Explanation:

1) **Sleep myoclonus/hypnic jerk** is an involuntary twitch that occurs just as a person is beginning to fall asleep

2) **Parkinsonism** is associated with antipsychotic medication resulting in stiffness, bradykinesia and tremor

3) **Athetosis** is slow involuntary writing movements of the limbs caused by degeneration of the basal ganglia, as seen in Huntington's disease. It can also be secondary to stroke or trauma.

Question 4

Correct

Mark 5.00 out of 5.00

Personality disorder

If all the general criteria for a personality disorder were met, what personality disorder would the traits described in these vignettes be indicative of?

1. Mr Whitaker is berated by some of his acquaintances for holding onto money very tightly. He claims he has to hold onto it for future catastrophies, and will never offer to pay for someone else's drink. He is regarded as miserly and uptight.

Anankastic personality disorder ✔

2. Liliana is often at the centre of any party, and cannot bear to be overlooked or ignored. She makes sure she's the centre of attention by wearing provocative clothes, laughing loudly and talking copiously and vivaciously.

Histrionic personality disorder ✔

3. Mike always seems to have several girlfriends, often at the same time. They find him attractive and seductive, and he impresses them with his stories of evading the law for his own profit. The relationships don't last long and Mike doesn't seem bothered when they come to an end. Antisocial personality disorder ✔

4. Nathan demonstrates the self-perception of being unique and superior to his peers.

Narcissistic personality disorder ✔

5. Mary is persistently unreliable at work and has poor control of her finances.

Borderline personality disorder- Emotionally unstable type. ✔

Explanation:

1) **Anankastic** personality disorder: Traits include a preoccupation with orderliness and control (including over money), at the expense of flexibility. Rules are very rigidly maintained.

2) **Histrionic** personality disorder: Traits include excessive attention-seeking emotions including inappropriately seductive behaviour.

3) **Antisocial** personality disorder: Traits include ease on forming relationships, but not keeping them, superficial charm, lack of remorse for criminal or law-breaking activity, disregard for the rights of others.

4) **Narcissistic** personality disorder can be recognised by elevated sense of importance and significance compared to others.

5) **Emotionally unstable** personality disorder usually involves marked impulsivity and a desultory approach to life; this usually leads to difficulties with finances and at work.

Question 5

Correct

Mark 3.00 out of 3.00

Diagnosis of acute disturbances
In each of the below scenarios, which of the following options would be the most likely diagnosis?

1. Mrs. Greaves is a 74-year-old woman who has recently had an elective hip replacement. Whilst on the orthopaedic ward she develops confusion, dyspnoea and chest pain 3 days postoperatively. Pulmonary embolism ✔

2. Helen Honey is a 6-year-old girl whose parents take her to the GP after they and her teachers notice that she is exhibiting episodes of blank staring and unresponsiveness lasting 10 to 15 seconds, which Helen is unaware of. Absence seizures ✔

3. Mrs. Ives is a 32-year-old woman with a diagnosis of bipolar affective disorder. She develops coarse tremor and ataxia after going for a run whilst staying at her holiday let in Spain during her summer holiday. Lithium toxicity ✔

Explanation:

1) Chest pain and dyspnoea would make a **pulmonary embolus** more likely than the other options. Significant hypoxia could lead to confusion.

2) The second scenario appears to describe **absence seizures** (also known as petit mal seizures) with classic brief periods of unresponsiveness but without any movement abnormalities.

3) The third example describes likely **lithium toxicity** due to dehydration. The patient has exerted themselves in a warm environment and has experienced fluid loss as a result.

Question 6

Correct

Mark 3.00 out of 3.00

Anxiety disorders

In each of the below scenarios, which of the following options would be the most likely diagnosis?

1. Matthew attends a CAMHS clinic. He has recently changed schools due to experiencing several difficulties. He also refuses to go out with friends or parents because he finds it uncomfortable to eat in front of people. Social phobia ✔

2. Beth is a 22-year-old who has been experiencing a two-week history of low mood, nervousness, poor concentration and hopelessness following the breakdown of her relationship with her boyfriend three months ago. Adjustment disorder ✔

3. Lauren is a 15-year-old who attends CAMHS clinic following a change in behaviour; her parents describe that she often blinks repeatedly or counts on her fingers. Lauren eventually confides that she repeatedly sees horrible images in her head.

Obsessive-compulsive disorder ✔

Explanation:

1) Matthew has a diagnosis of social phobia or social anxiety characterised by difficulties in a school or social environment.

2) Beth is suffering from adjustment disorder, which occurs up to 3 months (usually within 1 month) after a stressor and does not persist for more than 6 months after the stressor.

3) Lauren has a diagnosis of obsessive-compulsive disorder describing obsessional intrusive thoughts and counting compulsions.

Question 7

Partially correct

Mark 4.00 out of 5.00

Gait disorders

Which of the following disorders are mostly likely to be the diagnosis for each of the below scenarios?

1. Mrs Abigail Davidson is a 56-year-old woman who has been experiencing difficulties in starting to walk although she has little difficulty with movements involving her arms. She has no tremors of note. Lower motor neurone lesion ✖

2. Mr Daniel Gates is a 19-year-old man who has long-standing difficulties with mobility. He walks in a semi-crouched state and his lower legs tend to cross over each other as he walks. Upper motor neurone lesion ✔

3. Mr George Parker is a 45-year-old man who has been experiencing a gradual deterioration in his ability to walk. His steps are chaotic and variable in length and impaired by uncontrolled movements of his limbs at points. Huntington's disease ✔

4. Mrs Susan Bellows is a 72-year-old woman who has long-standing difficulties with mobilising. She walks bent forward and sometimes takes numerous small steps, seemingly to catch up with her body's momentum. Parkinson's disease ✔

5. Mrs Violet Walters is a 68-year-old woman who presents to her GP with a sudden onset difficulty in walking that started 2 days ago. She seems to be uncertain in her gait and places her feet far apart when taking a stride. She cannot stand with her feet together.

Cerebellar dysfunction ✔

Explanation:

1) Difficulty in gait initiation is typical in **normal pressure hydrocephalus**. It can also occur in Parkinsonism but difficulty in initiating movements tends to occur similarly in upper and lower limbs.

2) A gait in which the legs cross over or scissor is typical of an upper motor neurone lesion, particularly in **cerebral palsy.**

3) A chaotic gait impaired by choreic movements is characteristic of **Huntington's disease.**

4) A festinate gait is characterised by rapid small steps done in an attempt to keep the centre of gravity in between the feet while the trunk is leaning forward involuntarily and is classical in **Parkinson's disease** and similar disorders.

5) This scenario describes a broad-based gait and symptoms suggestive of truncal ataxia, both of which are associated with **cerebellar disease.**

Correct Answer is :

1. Mrs Abigail Davidson is a 56-year-old woman who has been experiencing difficulties in starting to walk although she has little difficulty with movements involving her arms. She has no tremors of note. - Normal pressure hydrocephalus

2. Mr Daniel Gates is a 19-year-old man who has long-standing difficulties with mobility. He walks in a semi-crouched state and his lower legs tend to cross over each other as he walks. - Upper motor neurone lesion

3. Mr George Parker is a 45-year-old man who has been experiencing a gradual deterioration in his ability to walk. His steps are chaotic and variable in length and impaired by uncontrolled movements of his limbs at points. - Huntington's disease

4. Mrs Susan Bellows is a 72-year-old woman who has long-standing difficulties with mobilising. She walks bend forward and sometimes takes numerous small steps, seemingly to catch up with her body's momentum. - Parkinson's disease

5. Mrs Violet Walters is a 68-year-old woman who presents to her GP with a sudden onset difficulty in walking that started 2 days ago. She seems to be uncertain in her gait and places her feet far apart when taking a stride. She cannot stand with her feet together. - Cerebellar dysfunction

Question 8

Correct

Mark 3.00 out of 3.00

Receptor mechanisms (1)
Using the definitions below, identify the terminology best suited:

This refers to the amount of the drug needed to produce a particular effect compared to another standard drug with similar receptor profile	Potency ✔
This refers to the ability of the drug to bind to its appropriate receptor	Affinity ✔
This refers to how well the drug produces the expected response	Efficacy ✔

Your answer is correct.

Potency of a drug with receptor binding action refers to the amount of the drug needed to produce a particular effect compared to another standard drug with similar receptor profile ('vigor').

Affinity refers to the ability of the drug to bind to its appropriate receptor ('affection').

Efficacy refers to how well the drug produces the expected response i.e. the maximum clinical response produced by a drug ('productivity').

Haloperidol is more potent than chlorpromazine - 5 mg of haloperidol can achieve the same effect as 100 mg of chlorpromazine. These drugs, however, produce equivalent maximal clinical response i.e. equally efficacious but not equipotent.

The correct answer is: This refers to the amount of the drug needed to produce a particular effect compared to another standard drug with similar receptor profile → Potency, This refers to the ability of the drug to bind to its appropriate receptor → Affinity, This refers to how well the drug produces the expected response → Efficacy

Question 9

Correct

Mark 4.00 out of 4.00

Class of Antipsychotics

Which of the following antipsychotics match with each of the below chemical compound types?

1. Phenothiazine Chlorpromazine ✔

2. Butyrophenone Haloperidol ✔

3. Dibenzodiazepine Clozapine ✔

4. Dibenzothiazepine Quetiapine ✔

Explanation: 1) Chlorpromazine, prochlorperazine, promazine, thioridazine and fluphenazine are examples of phenothiazine compounds. 2) Haloperidol, droperidol and benperidol are examples of butyrophenones. 3) Clozapine is the main dibenzodiazepine. 4) Quetiapine is the main dibenzothiazepine.

Correct Answer is :

1. Phenothiazine - Chlorpromazine

2. Butyrophenone - Haloperidol

3. Dibenzodiazepine - Clozapine

4. Dibenzothiazepine - Quetiapine

Question 10

Correct

Mark 5.00 out of 5.00

Mechanism of action II
Regarding the pharmacodynamics of the above drugs:

1. Which ONE of the drugs is a partial D2 agonist? Aripiprazole ✔

2. Which ONE of the drugs is a 5-HT1A agonist? Buspirone ✔

3. Which ONE of the antipsychotics binds to D1 receptors? Olanzapine ✔

4. Which ONE of the antipsychotics binds with high affinity to 5-HT2A receptors? Olanzapine ✔

5. Which ONE of the drugs is a multimodal agent that acts on 5-HT1A autoreceptor as well as the serotonin transporter? Vortioxetine ✔

Explanation:

1) From a pharmacological perspective, aripiprazole is different from other antipsychotic agents, as it is the only approved antipsychotic that reduces dopaminergic neurotransmission through D2 partial agonism, not D2 antagonism.

2) Buspirone is a full agonist at presynaptic 5-HT1A receptors and a partial agonist post-synaptically.

3 + 4) Olanzapine binds with high affinity to the following receptors: serotonin 5HT2A/2C, 5HT6, dopamine D1-4, histamine H1, and adrenergic a1 receptors.

5) Vortioxetine is a relatively new antidepressant that acts on the serotonin transporter (SERT) to reduce reuptake, in line with SSRIs. In addition, it also acts as an agonist of 5-HT1A on the presynaptic neuron, in line with pindolol and buspirone. This prevents the desensitization of this autoreceptor, and thus accelerates the antidepressant effects.

Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4296590/

Correct Answer is :

1. Which ONE of the drugs is a partial D2 agonist? - Aripiprazole

2. Which ONE of the drugs is a 5HT1A agonist? - Buspirone

3. Which ONE of the antipsychotics binds to D1 receptors? - Olanzapine

4. Which ONE of the antipsychotics binds with high affinity to 5HT2A receptors? - Olanzapine

5. Which ONE of the drugs is a multimodal agent that acts on 5HT1A autoreceptor as well as the serotonin transporter? - Vortioxetine

Question 11

Correct

Mark 4.00 out of 4.00

Neurohistology 2

Each of the cell types in the central and peripheral nervous systems have particular characteristics. Please match the following cell types to the descriptions below.

1. Which cell types commonly use glutamate as a neurotransmitter? (Please choose TWO answers) Pyramidal cells ✔

Granular cells ✔

2. Which cell type transmits motor signal from the cortex to the spinal cord? Pyramidal cells ✔

3. Which cell type lines the cerebral ventricles? Ependymal cells ✔

Explanation:

1) Numerous neurons utilise glutamate as a neurotransmitter, including **granular** and **pyramidal** cells. Glutamate is also shuttled via astrocytes, though these cells do not use glutamate as a neurotransmitter primarily.

2) **Pyramidal** cells, particularly Betz cells, are part of the corticospinal tracts that conveys motor signals from the motor cortex to the spinal cord.

3) **Ependymal** cells line the cerebral ventricles.

Question 12

Correct

Mark 4.00 out of 4.00

Neuropharmacology

Medications used in psychiatric practice often cause side effects, either because of their effects on target receptors in different areas of the body or affinity to non-target receptors. Which of the following receptors are associated with the below side-effects?

1. Mrs Oliver is a 32-year-old woman taking amisulpride who attends clinic and informs you that her periods have stopped.

D2 receptors ✔

2. Mr Pierce is a 52-year-old man who has started a new medication but describes episodes of dizziness in the mornings or when standing up. Alpha-1-adrenergic receptors ✔

3. Mr Radford is a 68-year-old man who has started taking an antidepressant he had many years ago. He attends his GP reporting difficulty-voiding urine. Muscarinic receptors ✔

4. Mrs Stevens is a 45-year-old woman who has been on a medication for 6 weeks but is keen to change to an alternative as she finds she is sleeping excessively and finds it difficult to wake in the mornings. Histamine receptors ✔

Explanation:

1) Most antipsychotics **block D2 receptors** and have the potential to cause hyperprolactinaemia due to blockade in the pituitary. This is more frequent in patients taking amisulpride, risperidone, paliperidone or a first generation antipsychotic. Common consequences of this are infertility and osteoporosis in the long-term. Galactorrhoea is also possible in both men and women.

2) The second scenario appears to describe episodes of postural hypotension which can be linked to antagonism of **alpha-1-adrenergic** receptors. TCAs, mirtazapine and some antipsychotics are associated with this effect.

3) **Antimuscarinic** effects, which are common side-effects of TCAs, can cause urinary retention, as well as dry mouth, blurred vision and a variety of other effects.

4) **Histamine antagonism**, such as that caused by mirtazapine, can cause marked sedation, which some patients find undesirable. Promethiazine and quetiapine also have this effect.

Question 13

Correct

Mark 4.00 out of 4.00

Function of Neuroglial cells
Identify the Neuroglial cells with their functions;

1. Cells which provide myelination to central nervous system — Oligodendrocytes ✔

2. Cells which provide myelination to peripheral nervous system — Schwann cells ✔

3. Cells which transport glutamate to promote clearance from synapse — Astrocyte ✔

4. Cells which line the ventricles of the brain and the central canal of spinal cord — Ependymal cells ✔

Explanation:

Oligodendrocytes provides myelination to central nervous system and **Schwann** cells provides myelination to peripheral nervous system.

Rapid removal of glutamate from the synaptic space is required for the survival of releasing neurons. Although glutamate transporters are expressed by all CNS cell types, **astrocytes** are the cell type primarily responsible for glutamate uptake.

Ependymal cells line the CSF filled ventricles in the brain and the central canal of spinal cord.

Ref: https://www.ncbi.nlm.nih.gov/pubmed/10975906

Question 14

Correct

Mark 5.00 out of 5.00

Theme- Memory
Match the following descriptions with the correct type of memory described

1. Remembering how to eat breakfast. Procedural memory ✔

2. Remembering what you ate for breakfast. Declarative memory ✔

3. Remembering what breakfast is. Semantic memory ✔

4. Remembering after you have eaten breakfast that you need to pick up more cereal on your way home from work.
Prospective memory ✔

5. Remembering that after breakfast you took your bike out and cycled your usual route to work, and this felt tedious.
Episodic memory ✔

Explanation:

1) **Procedural** memory, or 'knowing how', is the unconscious memory of skills and how to do things. It is acquired through repetition and practice.

2) **Declarative** memory is memory of facts and events that can be consciously recalled (also known as explicit memory).

3) **Semantic** memory is a structured record of meanings, concepts, facts, knowledge about the world; it's to do with general factual knowledge that is shared with others, independent of personal experience.

4) **Prospective** memory refers to that which needs to be remembered in the future, and may be defined as 'remembering to remember'; it can be either event-based or time-based, and is often triggered by a cue (e.g. finishing breakfast).

5) **Episodic** memory refers to experiences and specific events in time in serial form, from which we can reconstruct the actual events that took place in our lives; it is the memory of autobiographical events, and has an emotional charge.

Question 15

Correct

Mark 1.00 out of 1.00

Which of the following defence mechanisms is most prominent in subjects with narcissism?

Select one:

- Idealisation
- Discrimination
- Acting out
- Anticipation
- Projection ✔

Your answer is correct.

Projection is commonly seen in narcissism. It involves taking one's own unacceptable qualities or feelings and ascribing them to other people. For example, a narcissist with a strong dislike for someone, might instead believe that the other person does not like him. Narcissistic defences include the following:

1. repression
2. denial
3. rationalisation
4. projection

The correct answer is: Projection

Question 16

Correct

Mark 1.00 out of 1.00

Avoidance is a defence mechanism most commonly used in:

Select one:

- OCD
- Agoraphobia ✔
- Hypochondriasis
- Somatoform disorder
- Depression

Your answer is correct.

Avoidance is a powerful cognitive and behavioural strategy that serves as a negative reinforcer to reduce anxiety associated with **agoraphobia**.

(Most questions on defence mechanism/cognitive styles are based on psychotherapeutic theoretical grounding; not on quantitative evidence regarding frequency and prevalence)

F40.0 Agoraphobia include fears not only of open spaces but also of related aspects such as the presence of crowds and the difficulty of immediate easy escape to a safe place (usually home). The term therefore refers to an interrelated and often overlapping cluster of phobias embracing fears of leaving home: fear of entering shops, crowds, and public places, or of travelling alone in trains, buses, or planes.

The correct answer is: Agoraphobia

Question 17

Correct

Mark 1.00 out of 1.00

Which of the following is true regarding adult personality over time?

Select one:

- Males become more introverted with age ✔
- Neuroticism increases with age in men
- Females show a significant decline in psychoticism but not neuroticism with age
- Extraversion increases with age in men
- Males show a significant increase in psychoticism with age

Your answer is correct.

Males and females both show a decrease in extraversion, neuroticism and psychoticism with ageing.

Ref: Pedersen NL, Reynolds CA. Stability and change in adult personality: Genetic and environmental components. European Journal of Personality. 1998 Sep;12(5):365-86.

The correct answer is: Males become more introverted with age

Question 18

Correct

Mark 1.00 out of 1.00

In Piaget's model of cognitive development, in which of the following stages does 'conservation' appear?

Select one:

- Preoperational stage
- Sensorimotor stage
- Concrete operational stage ✔
- Formal operational stage
- Proximal development

Your answer is correct.

Lack of conservation, presence of centration, irreversibility, class inclusion, and transitive inference are all characteristics of preoperative thought. The preoperational stage lasts from around 2-7 years of age.

Most children learn the rules for conservation at **concrete operational stage** (age 7 onwards).

The correct answer is: Concrete operational stage

Question 19

Correct

Mark 1.00 out of 1.00

Children often go through a process of using an item or non-physical object as a buffer between fantasy and reality during periods of anxiety. Which of the following best describes this phenomenon?

Select one:

- Transitional object ✔
- Splitting
- Parallel process
- Imprinting
- Latency

Your answer is correct.

Infants initially have a period of so-called subjective omnipotence as their desires are fulfilled seemingly automatically, despite the objective reality being different. Realising that they are separate to their mothers or other caregivers can be distressing for the infant as they are reliant on another for their needs. Through fantasizing about the object of its wishes the child will find comfort. A transitional object can be used in this process. Donald Winnicott described this phenomenon.

Winnicot observed that the transitional object may be any material object (often a blanket, piece of cloth or a soft toy) to which young children frequently develop intense, persistent attachments. This object would be manipulated, sucked, or stroked, and often became an indispensable aid for falling asleep.

The correct answer is: Transitional object

Question 20

Correct

Mark 1.00 out of 1.00

Rapprochement refers to a development stage in Mahler's separation-individuation theory, which occurs between the ages of:

Select one:

- 14-24 months ✔
- 4-6 months
- 6-8 months
- 10-12 months
- 36-48 months

Your answer is correct.

Rapprochement refers to a development stage in Mahler's separation-individuation theory, which occurs between 14-24 months. The child uses the caregiver as a base for exploration and frequently returns to them for reassurance whilst exploring their environment.

The correct answer is: 14-24 months

Question 21

Correct

Mark 1.00 out of 1.00

In Margaret Mahler's Separation-Individuation Theory, at what age is a child in the Object Constancy Subphase?

Select one:

- 2 years – 5 years ✔
- 10 months – 18 months
- 2 months – 5 months
- 5 months – 10 months
- 18 months – 24 months

Your answer is correct.

Object constancy: (24+ months) In Margaret Mahler's Separation-Individuation Theory successful completion of this phase marks the development of an internalized mental model of the mother, which unconsciously accompanies and supports the child even when they are physically separated. In addition, a sense of individuality begins to develop. The degree of ambivalence in the internalized model implicates the formation of a healthy self-concept and self-confidence.

The correct answer is: 2 years – 5 years

Question 22

Correct

Mark 1.00 out of 1.00

Hagir Ahmed is struggling with her two sons, aged 14 and 19, who are proving to be strong-willed and determined young men. Hagir perceives a difference between the behaviours and mindset of her 14-year-old compared with that of her 19-year-old. She wonders about the effect of puberty of their developing brains, and what sorts of changes occur in the brain, that distinguish adolescence from adulthood. Which of the following statements regarding brain changes from adolescence to adulthood is correct?

Select one:

- Decrease in grey matter and increase in white matter ✔
- No change
- Increase in both grey and white matter
- Increase in grey matter and decrease in white matter
- Decrease in both grey and white matter

Your answer is correct.

Between adolescence and adulthood the brain undergoes synaptic pruning, resulting in a steady decline of grey matter.

Myelogenesis is active during childhood and adolescence, peaking in the third decade of life, then decreases. (Ref: Hum Brain Mapp 2010;31(6):926-933 and Neuropsychiatr Dis Treat 2013;9:449-461)

The correct answer is: Decrease in grey matter and increase in white matter

Question 23

Correct

Mark 1.00 out of 1.00

You are preparing a lecture on personality disorder for medical students, and you come across studies involving newborn rhesus monkeys, used by Harlow in the 1950s and 1960s in order to illustrate and contribute to the development of which of the following psychological concepts?

Select one:

- Attachment ✔
- Hypersocialisation
- Projection
- Avoidance
- Starvation

Your answer is correct.

Harlow studied the mechanism by which newborn rhesus monkeys bond with their mothers, suggesting that attachment develops as a result of tactile comfort, and not just the provision of food.

The correct answer is: Attachment

Question 24

Correct

Mark 1.00 out of 1.00

Ainsworth constructed a strange situation experiment with:

Select one:

- 1 separation and 3 reunion episodes
- 2 separation and 3 reunion episodes
- 2 separation and 2 reunion episodes ✔
- 3 separation and 1 reunion episodes
- 1 separation and 2 reunion episodes

Your answer is correct.

Ainsworth constructed a strange situation experiment with **2 separation and 2 reunion episodes**. It is a controlled observation carried out in a comfortable room equipped with toys. An infant is observed in the presence and absence of its mother and a stranger in the vicinity in seven different combinations.

According to the infant's behaviour it is classified as type A, B or C.

A- Anxious avoidant

B- Secure

C- Anxious resistant

The correct answer is: 2 separation and 2 reunion episodes

Question 25

Correct

Mark 1.00 out of 1.00

Gender identity in a normal child develops around

Select one:

- 5-6 years of age
- 11-12 years of age
- 6-7 years of age
- 3-4 years of age ✔
- 1-2 years of age

Your answer is correct.

Gender identity refers to the unshakable self-conviction of being male or female that begins around 18 months and gets fixed by 24 to 30 months. By age 3, most children are able to label themselves as either a boy or a girl. Most children develop stable sense of gender identity by age 4.

The correct answer is: 3-4 years of age

Question 26

Incorrect

Mark 0.00 out of 1.00

At what age can a child copy a circle with crayon?

Select one:

- 6-12 months
- 3-4 years ✖
- 4-5 years
- 1-2 years
- 2-3 years

Your answer is incorrect.

Age 2-3: In terms of fine motor skills, a child can copy a straight line with a crayon.

By **30 months (2.5 years)**, he or she tries to copy a circle and a plus sign.

The correct answer is: 2-3 years

Question 27

Correct

Mark 1.00 out of 1.00

The range of pubertal time period in a boy lasts for duration of:

Select one:

- 5-6 years
- 6-7 years
- 7-8 years
- 2-3 years ✔
- 3-5 years

Your answer is correct.

Boys enter puberty at an average age of 11-12 (one year later than girls) but this may occur anytime.

One of the largest cohorts of the pubertal adolescents reported so far has estimated the duration of puberty in boys to be around **2.4 years**; in girls about 1.9 years. In this study, duration was defined by the difference in ages at onset of a pubertal growth spurt (OGS) and peak height velocity (PHV). Depending on the method used, the duration of puberty varies between 2 to 3 years in both sexes, with boys having longer duration than girls consistently (refer to Tanner's method and Gasser's method for further discussion).

The correct answer is: 2-3 years

Question 28

Correct

Mark 1.00 out of 1.00

Which of the following is INCORRECT with regards to child development?

Select one:

- A 2-year-old child can say more than 50 words
- A 5-year-old child can tell time by quarter of an hour accuracy ✔
- A 12-month-old child can speak up to 3 words
- A child starts babbling at 4 months of age
- A 4-year-old child speaks with correct grammar

Your answer is correct.

Reception class children (ages 4-5) should be beginning to read the time to the hour.

Year 1 children (ages 5-6) should be able to read the time to the hour or the half hour on analogue clocks.

Year 2 children (ages 6-7) should be able to read the time to the hour, half hour or quarter hour on analogue clocks.

The correct answer is: A 5-year-old child can tell time by quarter of an hour accuracy

Question 29

Correct

Mark 1.00 out of 1.00

When children are hospitalised for physical health problems, a short period of separation from their primary care giver ensues. This temporary separation from their loved one can cause:

Select one:

- Anaclitic depression ✔
- Privation
- Deprivation
- Pathological separation anxiety
- Childhood grief reaction

Your answer is correct.

Anaclitic depression or hospitalism

Spitz described this. When children are hospitalised for physical health problems, a short period of separation from primary caregiver ensues; this loss produces anaclitic (object loss) depression. It is thought to be counterproductive to a child's development. But recovery can be very good if maternal deprivation is kept to a minimum i.e. less than three months. To some extent, surrogate mothering can help an infant experiencing anaclitic depression.

The correct answer is: Anaclitic depression

Question 30

Correct

Mark 1.00 out of 1.00

A 4-year-old child asks 'When do clouds sleep, papa?' Which of the following features is she exhibiting?

Select one:

- Semiotic thinking
- Lack of conservation
- Animism ✔
- Conservation
- Egocentricity

Your answer is correct.

Animism refers to the attribution of life and consciousness to certain inanimate objects.

The correct answer is: Animism

Question 31

Correct

Mark 1.00 out of 1.00

Mahler is associated with:

Select one:

- Autistic phase ✔
- Conventional morality
- Anal phase
- Operational stage
- Individuality vs. inferiority

Your answer is correct.

Margaret Mahler proposed a separation-individuation theory in which a normal autistic phase was described.

The correct answer is: Autistic phase

Question 32

Correct

Mark 1.00 out of 1.00

A psychology student is keen to know about the ICD-10 duration criteria for the diagnosis of mania. The episode should last at least:

Select one:

- 1 week ✔
- 2 days
- 2 weeks
- 1 day
- 4 weeks

Your answer is correct.

Both ICD-10 and DSM-4 specify the manic episode should last at least 1 week and the disturbance is sufficiently severe to impair social and occupational functioning.

The correct answer is: 1 week

Question 33

Correct

Mark 1.00 out of 1.00

The symptoms of adjustment disorder resolve within:

Select one:

- 6 months of termination of the stressor ✔
- 1 week of termination of the stressor
- 2 weeks of termination of the stressor
- 1 month of termination of the stressor
- 3 months of termination of the stressor

Your answer is correct.

Adjustment disorder requires a stressor and that the symptoms should resolve within 6 months of termination of the stressor or its consequences.

The correct answer is: 6 months of termination of the stressor

Question 34

Correct

Mark 1.00 out of 1.00

What is the most common psychiatric diagnosis among those presenting to emergency departments following self-harm?

Select one:

- Adjustment disorder ✔
- Schizophrenia
- Post-traumatic stress disorder
- Depression
- Panic disorder

Your answer is correct.

The most common psychiatric diagnosis among those presenting to emergency departments following self-harm is adjustment disorder.

(Ref: Taggart et al; Gen Hosp psychiatry 2006: 28: 330-5)

The correct answer is: Adjustment disorder

Question 35

Correct

Mark 1.00 out of 1.00

Which of the following descriptions is suggestive of dissocial personality disorder?

Select one:

- Evidence of childhood conduct disorder ✔
- Sensitive to criticisms
- Emotionally aloof
- Flattened affectivity
- Unclear identity

Your answer is correct.

A callous lack of concern for others, irresponsibility, irritability, aggression, tendency to act outside social norms, a disregard for the feelings of others, inability to maintain enduring relationships, low threshold for violence, a tendency to blame others, evidence of childhood conduct disorder are the reported features of dissocial personality.

The correct answer is: Evidence of childhood conduct disorder

Question 36

Correct

Mark 1.00 out of 1.00

ICD-10 diagnostic criteria for dissocial personality disorder includes:

Select one:

- Low frustration tolerance ✔
- Diagnosis of conduct disorder
- Incapacity to form relationships
- Accepting inappropriate blame
- Persistent irritability

Your answer is correct.

ICD-10 diagnostic criteria include:

- a callous lack of concern for the feelings of others
- an incapacity to maintain (not establish) enduring relationships
- an incapacity to experience guilt
- a persistent proneness to blame others
- disregard for social norms
- a very low frustration tolerance

Conduct disorder during childhood and adolescence, though not invariably present, may further support the diagnosis.

Persistent irritability may be an associated feature.

The correct answer is: Low frustration tolerance

Question 37

Correct

Mark 1.00 out of 1.00

A HIV carrier with ring-like lesions on MRI scan is most likely to have a diagnosis of:

Select one:

- Pneumocystosis carinii
- Cytomegalovirus
- Tuberculosis
- Crytoptococcus
- Toxoplasmosis ✔

Your answer is correct.

Cerebral toxoplasmosis commonly affects patients with advanced HIV immunodeficiency. Toxoplasmosis with CNS involvement in patients who are immunocompromised can be severe and debilitating, and the condition may be fatal. Magnetic Resonance Imaging (MRI) scan shows multiple ring-enhancing lesions with extensive surrounding oedema in supratentorial as well as the infratentorial region.

The correct answer is: Toxoplasmosis

Question 38

Correct

Mark 1.00 out of 1.00

Which one of the following tests would be best used for testing executive function?

Select one:

- Thematic Apperception Test
- Minnesota Multiphasic Personality Inventory
- Trail Making Tests ✔
- Brief Psychiatric Rating Scale
- Halstead-Reitan Battery

Your answer is correct.

The Trail Making Test (TMT) has two parts and the times taken to complete each part are used to measure **central executive functioning**.

In **Part A**, the participant must draw a line to connect consecutive numbers, 1 to 25. This tests **visual screening, processing speed, attention**.

In **Part B** the participant connects numbers and letters in an alternating progressive sequence (i.e. connect A-1-B-2-C-3 and so on). This tests **central executive processes** of **task-set inhibition, cognitive flexibility**, and **the ability to maintain a response set**.

In order to measure central executive functioning, the difference in time taken to complete TMT-B and TMT-A (which has little executive input) is calculated.

The correct answer is: Trail Making Tests

Question 39

Correct

Mark 1.00 out of 1.00

A doctor, during a clinical interview, tells a patient, "I think I now have a good idea of your financial situation; perhaps we should talk about your family problems now". Which technique is the doctor utilising?

Select one:

- Empathy
- Repetition
- Facilitation
- Acknowledgement
- Transition ✔

Your answer is correct.

Transition is an interview technique used when moving from one to another topic. It is a very useful interview skill and preferably must be done as smoothly as possible to keep the patient interested.

The correct answer is: Transition

Question 40

Correct

Mark 1.00 out of 1.00

What is the most common finding in a newly diagnosed patient with Parkinson's disease?

Select one:

- Unilateral leg rigidity
- Unilateral resting tremor ✔
- Micrographia
- Postural instability
- Slowing of movement

Your answer is correct.

Unilateral resting tremor is the most common presenting feature in Parkinson's disease.

Ref: Adler and Ahlskog: Parkinson's disease and movement disorders Pg 72

The correct answer is: Unilateral resting tremor

Question 41

Correct

Mark 1.00 out of 1.00

A patient is suspected to have temporal lobe damage resulting from a vascular lesion. Which of the following visual signs may be elicited on perimetry?

Select one:

- Bitemporal hemianopia
- One-sided blindness
- Homonymous hemianopia
- Lower quadrantanopia
- Upper quadrantanopia ✔

Your answer is correct.

Temporal lesions produce upper quadrantanopia due to damage to optic radiation.

The correct answer is: Upper quadrantanopia

Question 42

Correct

Mark 1.00 out of 1.00

A patient breaks down and cries during interview. Which of the following technique is useful at this stage?

Select one:

- Ask a question with multiple choices
- Validation ✔
- Transition to different topic
- Distraction technique
- Avoid eye contact

Your answer is correct.

Validation statements facilitate the continuation of a clinical interview by 'normalising' experiences e.g. 'it is normal to cry at difficult times.'

The correct answer is: Validation

Question 43

Correct

Mark 1.00 out of 1.00

Mr. Y has a diagnosis of chronic schizophrenia. He has polyuria, polydipsia and becomes acutely confused. On further investigation, his serum sodium levels are notably low. What do you suspect?

Select one:

- Serotonin syndrome
- Dehydration
- Catatonia
- Neuroleptic malignant syndrome
- Water intoxication ✔

Your answer is correct.

Water intoxication occurs particularly in patients with chronic schizophrenia. The clinical presentation includes polyuria, polydipsia, hyponatraemia, confusion, seizures, cerebral and peripheral edema. Death can occur in extreme cases. The cause is generally unknown.

The correct answer is: Water intoxication

Question 44

Correct

Mark 1.00 out of 1.00

Which of the following is NOT a feature of late-onset schizophrenia?

Select one:

- Family history of schizophrenia is infrequent when compared to early onset
- Less pronounced formal thought disturbances
- White matter hyperintensities on MRI brain ✔
- Fewer negative symptoms
- Prominent hallucinations

Your answer is correct.

Late-onset schizophrenia is characterised by:

- a predominance of females
- fewer negative symptoms
- infrequent formal thought disorder
- prominent hallucinations (notably visual)

Associations include:

- social isolation
- sensory impairment

- premorbid paranoid personality traits.

Family loading for schizophrenia is lower than for early-onset cases (Howard et al., 2000).

White matter hyperintensities are not associated with late-onset schizophrenia (Howard 1995; Rivkin 2000) though one contradictory study exists favouring higher frequency of WMH in late-onset schizophrenia (Sachdev 1999).

The correct answer is: White matter hyperintensities on MRI brain

Question **45**

Correct

Mark 1.00 out of 1.00

A 44-year-old man with schizophrenia has intense suspiciousness and persecutory delusions. He tells you that his neighbours want him to prepare a home-made explosive with detonators. Which of the following would be the best response at this point of interview?

Select one:

- Perhaps we should ask your neighbours if this is true
- Why do you think your neighbours want you to do that? ✔
- No, I don't think this sort of thing could happen
- If you prepare explosives, we may have to involve the police
- It doesn't sound like a good idea to prepare explosives

Your answer is correct.

The main purpose of the interview is to collect clinically useful information that will enable diagnostic process and facilitate management. It is important to maintain a good therapeutic relationship with the patient while pursuing the above goals. Enquiring 'Why do you think your neighbours want you to do that?' serves both these purposes.

The correct answer is: Why do you think your neighbours want you to do that?

Question 46

Correct

Mark 1.00 out of 1.00

Regarding 'Quality Of Life-16', which of the following is true:

Select one:

- It is a self-administered scale ✔
- It is a structured interview
- It is a semi-structured interview
- The maximal score is 16
- It cannot be used in patients with chronic illnesses

Your answer is correct.

Flanagan's QOLS originally had 15 items, but was expanded to include a 16th item (independence). It measures five conceptual domains of quality of life:

1. material and physical well-being
2. relationships with other people
3. social, community and civic activities
4. personal development and fulfilment
5. recreation

It is most often used in samples with chronic illnesses. The QOLS is usually self-administered. It can also be completed by interview format. The QOLS can be completed in about 5 minutes. Scores can range from 16 to 112.

Ref: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC269997/

The correct answer is: It is a self-administered scale

Question 47

Correct

Mark 1.00 out of 1.00

A patient with schizophrenia often rocks back and forth. He is also observed to be thumping his head with his fist for no apparent reason. He is exhibiting which of the following symptoms?

Select one:

- Hemiballism
- Ambitendence
- Ambivalence
- Stereotypy ✔
- Mannerism

Your answer is correct.

This patient is demonstrating **stereotypy**; a symptom characterised by repetitive, functionless motor behaviour.

Stereotypies, mannerisms, and verbigeration are often elicited in patients with catatonic schizophrenia.

Hemiballism – hyperkinetic movement disorder characterised by violent involuntary proximal limb movements on one side of the body, caused by damage to the subthalamic nucleus of the basal ganglia. Unrelated to schizophrenia.

Ambitendency – a motor symptom where a patient who has been given an instruction alternates between automatic obedience and negativism e.g. when asked to shake hands, the patient appears indecisive, repeatedly extending and withdrawing their hand.

Ambivalence - described by Bleuler as one of the four primary symptoms of schizophrenia and was defined as having positive and negative emotions at the same time. Patients exhibiting ambivalence often have an inability to decide on an action.

Mannerisms - repetitive, goal-directed movements (e.g. saluting).

The correct answer is: Stereotypy

Question 48

Correct

Mark 1.00 out of 1.00

A patient that you see in your hospital corridor is moving his closed fist round to the right and round to the left in a repetitive manner. He is exhibiting:

Select one:

- Hemiballism
- Tonic-clonic seizures
- Mitgehen
- Stereotypy ✔
- Chorea

Your answer is correct.

Repetitive, non-goal-directed motor activity (e.g. finger-play, repeatedly touching, patting or rubbing self) is called **stereotypy**. It is an abnormality not inherent in act but in its frequency.

The correct answer is: Stereotypy

Question 49

Correct

Mark 1.00 out of 1.00

A man using mescaline 'smelt the noise of the trumpet'. This condition is known as:

Select one:

- Reflex hallucination
- Synaesthesia ✔
- Visceral hallucination
- Haptic hallucination
- Functional hallucination

Your answer is correct.

This is a classical example of **synaesthesia** described by Sims. The phenomenon of perceiving a stimulus of one modality in a different modality (may be single or multiple modalities) is called synaesthesia. E.g. tasting the music, hearing colours and smelling voices. It is not a hallucination as the perceived object has an appropriate stimulus.

Reflex hallucinations are a hallucinatory form of synaesthesia. A stimulus in one modality produces a hallucination in another modality.

The correct answer is: Synaesthesia

Question 50

Correct

Mark 1.00 out of 1.00

A psychologist asked Mr. Snowden whether he had any special powers or abilities that other people do not have. He replied that he wasn't sure as he never visited Russia. When asked to clarify, he asked, "If you throw twenty-and-two blackbirds, would you know where does the evening start?" This form of speech is called:

Select one:

- Circumstantiality
- Neologism
- Paraphasia
- Tangentiality ✔
- Verbigeration

Your answer is correct.

Tangentiality - speech is organised, but not goal directed. The subject answers a question in an indirect manner with vague relevance. The answer appears distant and drifted away from the original goal.

The correct answer is: Tangentiality

Question 51

Correct

Mark 1.00 out of 1.00

In psychopathology, the term 'moria' refers to:

Select one:

- Tendency to cry inappropriately
- Tendency to be inappropriately euphoric ✔
- Tendency to choke when drinking fluids
- Tendency to be restless and hyperactive
- Tendency to be aggressive

Your answer is correct.

Puerile, childish, frivolous, or silly behaviour occurs with right frontotemporal lesions and in frontotemporal dementia. In such cases, moria (foolish or silly euphoria) and Witzelsucht (a tendency to tell inappropriate jokes are seen).

The correct answer is: Tendency to be inappropriately euphoric

Question **52**

Correct

Mark 1.00 out of 1.00

The average capacity of short term memory is:

Select one:

- 7+/-2 ✔
- 9+/-2
- 5+/-2
- 3+/-2
- 6+/-2

Your answer is correct.

Short term memory:

It holds a small amount of information. The capacity of STM according to Miller is 7+/- 2, as evident while testing digit span.

By chunking, larger information could be sorted into 7+/- 2 items and effectively stored. Chunks are bits of information coordinated together with the help of long-term memory.

Unaided, STM lasts 15 to 30 seconds. By maintenance rehearsal (repetition of items in one's mind) this duration can be increased further up to indefinite periods.

STM largely uses acoustic coding (mostly) or visual coding. Recall of information is effortless and usually error-free.

Information is held in STM by the process of rehearsal.

Loss of information from STM occurs mainly through displacement (newly acquired items entering STM displaces existing material) and decay (recently acquired material has a higher trace strength than older items)

The correct answer is: 7+/-2

Question 53

Correct

Mark 1.00 out of 1.00

Which one among the following is a mode of retrieval from long-term memory?

Select one:

- Recall
- Relearning
- Recognition
- All of the above ✔
- Reintegration

Your answer is correct.

Modes of retrieval (LTM store being passed onto STM) are through:

- recognition (solving MCQs)
- recall (actively searching and reproducing)
- relearning
- confabulation
- reintegration (recollection of past experiences based on certain cues)

An eyewitness testimony is a reconstructive memory, which is a mode of retrieval from long-term memory. However, a reconstructive memory of events as in eyewitness testimony is affected by the type of questioning asked to elicit the information.

The correct answer is: All of the above

Question 54

Correct

Mark 1.00 out of 1.00

With regard to attention, which of the following is incorrect?

Select one:

- Divided attention is paying attention to multiple tasks at a time
- Concentration is sustained attention
- Stroop test is a test for selective attention
- Alternating attention is the highest level of attention. ✔
- Selective attention is the ability to avoid distractions

Your answer is correct.

Divided attention is the highest level of attention, and it refers to the ability to respond simultaneously to multiple tasks or multiple task demands.

Alternating attention refers to the ability to switch between tasks; to stop one task to participate in another and then be able to return to the initial task.

The correct answer is: Alternating attention is the highest level of attention.

Question 55

Correct

Mark 1.00 out of 1.00

Incorporating 'applied tension' is necessary when doing exposure therapy with what particular group of phobias?

Select one:

- Blood-injection-injury type phobia ✔
- Claustrophobia
- Agoraphobia
- Situational phobia
- Social phobia

Your answer is correct.

Blood-injection-injury type phobia is unique in that a paradoxical drop in blood pressure occurs during exposure to a particular stimulus, leading to fainting. Applied tension involves contraction of large muscle groups to maintain a normal blood pressure during exposure. This practice is contrary to the progressive muscle relaxation that is employed when treating other forms of anxiety.

The correct answer is: Blood-injection-injury type phobia

Question 56

Correct

Mark 1.00 out of 1.00

A 45-year-old female patient is undergoing regular chemotherapy for her breast cancer; at her last visit she became nauseated and vomited as she entered the clinic room. This is likely to be related to:

Select one:

- Positive reinforcement
- Classical conditioning ✔
- Modeling
- Negative reinforcement
- Habituation

Your answer is correct.

Classical conditioning occurs when a patient associates some aspect of the oncology clinic (e.g. the room, smell, the sight of the intravenous infusion bottle) with the nausea previously experienced during the intravenous process itself. Thus, the mere sight or smell of one or more components of the experience can produce the conditioned response of nausea and vomiting.

The correct answer is: Classical conditioning

Question 57

Correct

Mark 1.00 out of 1.00

A 45-year-old man was admitted to the acquired brain injury unit with severe memory impairment. Which of the common clinical tests is used in testing his immediate memory?

Select one:

- Recall of recent topics in news
- Recall of distant personal events
- Recall of items after 5 minutes
- Recall of last meal
- Digit span ✔

Your answer is correct.

The terms used in psychology and psychiatry are somewhat different which could lead to some confusion.

The terms used in psychology are short term memory (immediate memory in psychiatry) and long-term memory (recent memory and remote memory in psychiatry).

In psychology, short term memory refers to immediate memory, which is tested by the recall of digits immediately after their presentation (**digit span**).

However, in psychiatry it is often referred to as a test of immediate memory. Recollection of a name and an address after 5 minutes is seen as a test of recent memory (which, in fact, is a part of LTM and should not be considered as a test of STM)

The correct answer is: Digit span

Question 58

Correct

Mark 1.00 out of 1.00

According to Aaron Beck, depression results from:

Select one:

- High expressed emotions
- Aggression turned inwards
- Genetic risk
- Distorted negative thoughts ✔
- Low expressed emotions

Your answer is correct.

Aaron Beck first proposed the theory behind cognitive distortions that are associated with depression.

The correct answer is: Distorted negative thoughts

Question 59

Correct

Mark 1.00 out of 1.00

With regards to exposure-based treatment approaches, which of the following is true?

Select one:

- Exposure should be sufficiently long for response to occur ✔
- Direct exposure to feared object is detrimental
- An overtly safe environment should not be created
- Experience of anxiety during the first session must be avoided
- Patients must be discouraged from being exposed outside the treatment context

Your answer is correct.

Exposure therapy has been shown to be an evidence-based treatment component for phobias, GAD, OCD, panic disorder. There are several variations of exposure therapy (in vivo, graded, flooding, imaginal, interoceptive, systematic desensitisation). Over time, people find that their reactions to feared objects or situations decrease. Exposure can help weaken previously learned associations between feared objects, activities, or situations and bad outcomes.

The correct answer is: Exposure should be sufficiently long for response to occur

Question 60

Correct

Mark 1.00 out of 1.00

Two antidepressants A and B were introduced. Both A and B had similar mechanism of action and similar efficacy. A shows 50% failure rate. B shows 50% success rate. Drug B reported much better sales when marketed than drug A. Select the cognitive heuristic in this case.

Select one:

- Anchoring
- Simulation
- Representation
- Availability
- Framing ✔

Your answer is correct.

Framing is a cognitive heuristic in which people tend to reach conclusions based on the 'framework' within which a situation was presented. A judgment is made based on whether the options are presented with positive or negative connotations.

The correct answer is: Framing

Question 61

Correct

Mark 1.00 out of 1.00

Which brain region plays a predominant role in working memory?

Select one:

- Hippocampus
- Amygdala
- Occipital lobe
- Frontal cortex ✔
- Cerebellum

Your answer is correct.

Dorsolateral prefrontal cortex is considered to be a significant seat of working memory capacity. Impairment of fronto-parietal networks in schizophrenia is thought to be related to working memory deficits seen in this disorder.

The correct answer is: Frontal cortex

Question 62

Correct

Mark 1.00 out of 1.00

Which theory of perception is most aligned with the view that "the whole is more than the sum of the parts"?

Select one:

- Kohut
- Mahler
- Freud
- Adler
- Gestalt ✔

Your answer is correct.

When the perceptual system forms a percept or **gestalt**, the whole thing has a reality of its own, independent of the parts. The Gestalt psychologist Kurt Koffka made a famous statement about this: "The whole is other than the sum of its parts." This statement is often translated into English as, "The whole is greater than the sum of the parts." Koffka did not like that translation. He firmly corrected students who substituted "greater" for "other" (Heider, 1977). "This is not a principle of addition," he said. The statement as originally worded was supposed to mean that the whole had an independent existence in the perceptual system

(Excerpt from Dewey RA. The Whole is Other than the Sum of the Parts. Retrieved from http://www.intropsych.com/ch04_senses/).

The correct answer is: Gestalt

Question 63

Correct

Mark 1.00 out of 1.00

Motivational interviewing is associated with:

Select one:

- Inventiveness
- Increasing cognitive dissonance ✔
- Negative reinforcement
- Premack's principle
- Learned resourcefulness

Your answer is correct.

Motivational interviewing attempts to encourage people to take on responsibility for their own life and to deal with their problems themselves. The nature, principles and techniques of MI are, without exception, found to relate to one or more of the principles of **cognitive dissonance**. In creating a cognitive dissonance, however subtly or gently you may do that, you are inevitably directing the client to where, in many ways, you as the therapist want them to be, to get them to be open to change.

(Ref: "MOTIVATIONAL INTERVIEWING Preparing People to Change Addictive Behaviour" by W.R. Miller & S. Rollnick, The Guildford Press, New York, London, 1991)

The correct answer is: Increasing cognitive dissonance

Question 64

Correct

Mark 1.00 out of 1.00

Sternberg's triarchic theory is a theory of

Select one:

- Temperament
- Adolescent development
- Sociocultural adapatation
- Intelligence ✔
- Psychosexual development

Your answer is correct.

According to Sternberg's triarchic theory, **intelligence** has three components: analytical, creative, and practical.

The correct answer is: Intelligence

Question 65

Correct

Mark 1.00 out of 1.00

The Stanford-Binet test is a test of

Select one:

- Behaviour
- Personality
- Intelligence ✔
- Attitude
- Memory

Your answer is correct.

It is a test of **intelligence**. Around 50% of the population are said to have an IQ between 90 and 110.

In the Stanford-Binet test, IQ is calculated as follows:

IQ = Measured Age/Chronological Age multiplied by 100.

The correct answer is: Intelligence

Question 66

Correct

Mark 1.00 out of 1.00

Which of the following statements about Eysenck's three-factor theory is true?

Select one:

- Extraversion is related to autonomic lability
- Introversion is associated with autonomic lability
- Introverts have a more reactive reticular activating system ✔
- Extraverts have a more reactive reticular activating system
- Extraversion is related to cortisol levels

Your answer is correct.

In introverts, the ascending reticular activating system (ARAS) is over-aroused and excited. In extraverts, the ARAS is under-aroused and inhibited.

The correct answer is: Introverts have a more reactive reticular activating system

Question 67

Correct

Mark 1.00 out of 1.00

According to Yerkes-Dodson Law, the level of arousal is related to performance of an act by:

Select one:

- J-shaped curve
- U-shaped curve
- Straight line
- Inverted U-shaped curve ✔
- Inverted J-shaped curve

Your answer is correct.

Yerkes-Dodson Law: **An inverted U-shaped curve** relates the **level of arousal** with the **performance** of an act. Optimum arousal (moderate) is required for best performance; too low or too high arousal proves to be a hindrance and can lead to a poorer performance. The optimal level of arousal for best performance is also influenced by the nature of the task. A simple well-rehearsed task will require a higher level of arousal than more complex, less well-rehearsed tasks.

The correct answer is: Inverted U-shaped curve

Question 68

Correct

Mark 1.00 out of 1.00

Autonomy, beneficence, non-maleficence in medical ethics are described as:

Select one:

- Utilitarianism
- Teleology
- Enshrinement
- Principlism ✔
- Deontology

Your answer is correct.

Principlism refers to Childress and Beauchamp's four prima facie principles in ethics: autonomy, beneficence, non-maleficence, and justice.

The correct answer is: Principlism

Question 69

Correct

Mark 1.00 out of 1.00

In infants born prematurely, after which week of pregnancy are we legally able to resuscitate an infant irrespective of parental wishes?

Select one:

- 20 weeks
- 16 weeks
- 22 weeks
- 21 weeks
- 25 weeks ✔

Your answer is correct.

In the United Kingdom women have access to termination of pregnancy for maternal reasons until 24 weeks' completed gestation, but it is accepted practice for children born at or beyond 25 weeks' gestation to be treated according to the child's perceived best interests even if this is not in accordance with parental wishes.

(Retrieved from Campbell AV. Viability and the moral status of the fetus. Ciba Found Symp. 1985;115:228-43. http://www.ncbi.nlm.nih.gov/pubmed/3849416).

There is no legislation imposing an age limit in UK for resuscitating premature babies as of now. If this is a choice, this must be chosen.

(Boyle et al. Ethics of refusing parental requests to withhold or withdraw treatment from their premature baby. J Med Ethics.2004; 30: 402-405)

The correct answer is: 25 weeks

Question 70

Correct

Mark 1.00 out of 1.00

Social deprivation has no role in the aetiology of which of the following disorders?

Select one:

- Bipolar Disorder
- ADHD
- Depression
- Alzheimer's disease ✔
- Schizophrenia

Your answer is correct.

There is **no** significant causal association between increased risk of **Alzheimer's disease** (AD) and **socioeconomic status** (SES). There **is** an association between **low education** and increased **AD risk**, but this is not mediated by adult SES or socioeconomic mobility.

(Karp et al., 2003; http://aje.oxfordjournals.org/content/159/2/175)

The correct answer is: Alzheimer's disease

Question 71

Correct

Mark 1.00 out of 1.00

A 29-year-old man with a long history of schizophrenia and currently being effectively treated in the community decides to donate blood. His treating team feels he has the capacity to make decision with respect to this issue. Which of the following principle the treating team has adhered to?

Select one:

- Maleficence
- Non-maleficence
- Beneficence
- Autonomy ✔
- Equality

Your answer is correct.

Individual freedom is based on the principle of **autonomy** in modern medical ethics. It is the principle that a person should be free to make his or her own decisions.

The correct answer is: Autonomy

Question 72

Correct

Mark 1.00 out of 1.00

An Asian immigrant in England is observed to practice his religion and cultural traditions at home but adapts well especially at work with good degree of fluency in both English and his native language. This type of enculturation is called:

Select one:

- Melting Pot
- Bisection
- Assimiliation
- Accommodation
- Biculturalism ✔

Your answer is correct.

The acculturative process involves acquisition and retention as well as relinquishing attitudes/values and practices in both majority and minority populations that continues over several generations. The rate of change and the circumstances that influence it vary greatly, both between and within groups. For these reasons, studies of groups experiencing acculturative change often divide the groups by temporal experience into first, second, and third-generation immigrants. Families within such groups have been categorised as **traditional, transitional**, or **bicultural**.

- **Traditional** families are characterised as using their native tongues rather than English, living in ethnic enclaves, avoiding interaction with majority cultural institutions, and maintaining preimmigration values and behaviours.
- **Transitional** families are characterised by greater fluency in the language of the host culture and by children who are becoming familiar with the values and social behaviours of the dominant majority population through attendance at school and school-related activities.

- **Bicultural** families are defined as those with a high degree of language fluency in their native languages as well as English, economic stability, and residence in multiethnic settings. Biculturalism appears to be more adaptive and associated with minimal acculturation stress.

The correct answer is: Biculturalism

Question **73**

Correct

Mark 1.00 out of 1.00

Theories of social origins of depression is related to which one of the following pair of individuals?

Select one:

- Crow and Johnstone
- Ingvar and Franzen
- Cerletti and Bini
- Watson and Crick
- Brown and Harris ✔

Your answer is correct.

In 1978, as the result of two community surveys carried out in Camberwell between 1969 and 1975 of women aged 18 and 65 and a series of comparable psychiatric patients, **Brown and Harris** published their influential book, 'Social Origins of Depression'.

The correct answer is: Brown and Harris

Question 74

Correct

Mark 1.00 out of 1.00

What is the incidence of seizures in patients taking clozapine at a dose of more than 600mg/day?

Select one:

- 0.15
- 0.05 ✔
- 0.02
- 0.01
- 0.005

Your answer is correct.

Risk of seizures at clozapine dose:

<300 mg/d = 1% to 2%

300 to 600 mg/d = 3% to 4%

>600 mg/d = 5%

The correct answer is: 0.05

Question 75

Correct

Mark 1.00 out of 1.00

Neuroleptic malignant syndrome (NMS) is associated with a mortality rate of:

Select one:

- 15-20% ✔
- 5-10%
- 25-50%
- 1-2%
- 2-5%

Your answer is correct.

5-20% of patients on oral medication and up to 30% of patients on depot formulations who develop the syndrome will die from the condition if untreated.

The correct answer is: 15-20%

Question 76

Correct

Mark 1.00 out of 1.00

Which one among the following is not a common side effect of sodium valproate?

Select one:

- Agranulocytosis ✔
- Hair loss with curly regrowth
- Tremor
- Weight gain
- Thrombocytopenia

Your answer is correct.

Common side effects of sodium valproate include weight gain, nausea, vomiting, diarrhoea, sedation, tremor, thrombocytopenia and hair loss with curly regrowth.

Agranulocytosis is listed as a rare/very rare side effect.

Ref: BNF https://bnf.nice.org.uk/drug/sodium-valproate.html#sideEffects

The correct answer is: Agranulocytosis

Question 77

Correct

Mark 1.00 out of 1.00

Mr Z is a 25-year-old gentleman being treated for first episode schizophrenia. He was initially started on risperidone 2mg daily which was subsequently increased to 4mg by week 4. Although he was noted to show good improvement with his delusions and hallucinations, the nurses on the ward noticed he is generally restless, constantly pacing up and down, foot stamping when seated, rocking back and forth and fidgeting in his chair. The nurses wondered whether this could be a side effect of his medication. Which of the following side effects of treatment is he experiencing?

Select one:

- Neuroleptic malignant syndrome
- Akathisia ✔
- Tardive dyskinesia
- Restless leg syndrome
- Pseudoparkinsonism

Your answer is correct.

Akathisia is a subjective state of motor restlessness, which is seen in 20-25% of patients prescribed with anti-psychotic medication. It is an unpleasant state of inner restlessness where there is a strong desire or compulsion to move. It may become prominent as the dose of antipsychotic medication is increased. Akathisia can be mistaken for psychotic agitation and has been linked with suicide and aggression towards others.

The correct answer is: Akathisia

Question 78

Correct

Mark 1.00 out of 1.00

A significant prolactin elevating effect is seen with one of the following:

Select one:

- Ziprasidone
- Clozapine
- Quetiapine
- Aripiprazole
- Amisulpiride ✔

Your answer is correct.

Amisulpride, sulpiride, risperidone and paliperidone are particularly associated with hyperprolactinaemia.

Ref Maudsley Prescribing Guidelines. 13th ed. Page 38.

The correct answer is: Amisulpiride

Question 79

Correct

Mark 1.00 out of 1.00

The antidepressant most associated with QTc prolongation is:

Select one:

- Venlafaxine
- Paroxetine
- Mirtazapine
- Amitriptyline ✔
- Moclobemide

Your answer is correct.

The antidepressant class most associated with QTc prolongation is **tricyclic antidepressants** such as **amitriptyline**. Moclobemide, citalopram, and venlafaxine have a low effect on QTc prolongation. Mirtazapine and SSRIs (except citalopram) have no effect on QTc prolongation.

The correct answer is: Amitriptyline

Question 80

Correct

Mark 1.00 out of 1.00

Which of the following psychotropics has a tendency to reduce gastrointestinal transit with a resultant slow absorption when given orally?

Select one:

- Amitriptyline ✔
- Risperidone
- Diazepam
- Fluoxetine
- Lithium carbonate

Your answer is correct.

Amitriptyline, due to its atropine-like anticholinergic action, can slow GI transit with a resultant slow absorption. However, nearly 95% of the orally administered drug gets absorbed.

The correct answer is: Amitriptyline

Question 81

Correct

Mark 1.00 out of 1.00

Mrs. Y is a 28-year-old pregnant woman with severe depression. The ideal antidepressant treatment of choice would be:

Select one:

- Venlafaxine
- Reboxetine
- Paroxetine
- Fluvoxamine
- Amitriptyline ✔

Your answer is correct.

The antidepressants recommended for treatment of depression in pregnancy are amitriptyline, nortriptyline, imipramine and fluoxetine.

The correct answer is: Amitriptyline

Question 82

Correct

Mark 1.00 out of 1.00

Which drug increases the risk of bleeding in patients on SSRIs?

Select one:

- Paracetamol
- Aspirin ✔
- Naproxen
- Ibuprofen
- Diclofenac

Your answer is correct.

There is an increased risk of bleeding when SSRIs are taken with **antiplatelet** drugs such as low-dose **aspirin** or **clopidogrel** and non-steroidal anti-inflammatory drugs (**NSAIDs**). However, the risk is higher with **aspirin** than other NSAIDs. A large Canadian study found patients taking both aspirin and an SSRI had a 42% increased risk of a bleeding episode - gastrointestinal bleeding, hemorrhagic stroke or other forms of bleeding that required hospital admission - compared with those taking aspirin alone. (Labos et al., 2011: http://www.cmaj.ca/content/early/2011/09/26/cmaj.100912)

The correct answer is: Aspirin

Question 83

Correct

Mark 1.00 out of 1.00

A 33-year-old man is admitted to Psychiatric Intensive Care Unit with an acute relapse of schizophrenia. He is started on a typical antipsychotic medication after which his symptoms reduce in intensity. But since this new treatment was started, he complains of feeling restless and is seen pacing up and down a lot. When you interview him, he appears to be shuffling often and not able to stay for long in the interview room. If you want to prescribe propranolol to this patient, which of the following medical conditions will be a contraindication for the prescription?

Select one:

- Asthma ✔
- Obesity
- Hypertension
- Akathisia
- Elevated eosinophils

Your answer is correct.

Propranolol and other **beta-blockers** are recommended to treat **akathisia**, but beta-receptor blockade causes **bronchospasm**, and so are **contraindicated in asthma**.

The correct answer is: Asthma

Question 84

Correct

Mark 1.00 out of 1.00

Which of the following is not a feature of neuroleptic malignant syndrome?

Select one:

- Incontinence
- Bradycardia ✔
- Tremor
- Rigidity
- Diaphoresis

Your answer is correct.

Bradycardia is not a feature of neuroleptic malignant syndrome (NMS).

NMS can occur at anytime during treatment with neuroleptics. It consists of the tetrad of:

1. extreme hyperthermia
2. severe muscular rigidity
3. confusion
4. autonomic fluctuations (tachycardia and elevated/labile blood pressure)

Other clinical features of neuroleptic malignant syndrome include diaphoresis, dysphagia, tremor, shuffling gait, tachycardia, dyspnoea, incontinence, delirium progressing to lethargy, stupor, coma.

The correct answer is: Bradycardia

Question 85

Correct

Mark 1.00 out of 1.00

Which of the following is the most likely change seen in the ECG of patients taking lithium?

Select one:

- Flattened trace
- ST elevation
- Prolonged QRS
- Prolonged QTc
- Changes in T wave morphology ✔

Your answer is correct.

The most frequently observed change during treatment was **flattening or inversion of the T wave**.

- Heart rate decreased
- P-Q interval increased slightly during therapy
- There was no change in the ST segment
- There was no change in QTc

Well-monitored lithium treatment of patients with serum lithium concentrations of about 0.6 mmol/L does not cause serious ECG abnormalities. In the absence of symptoms or signs of heart disease, routine monitoring of ECG is not necessary during lithium treatment.

(Ref: Bucht G, Smigan L, Wahlin A, Eriksson P. ECG changes during lithium therapy. A prospective study. Retrieved from http://onlinelibrary.wiley.com/doi/10.1111/j.0954-6820.1984.tb03777.x/)

The correct answer is: Changes in T wave morphology

Question 86

Not answered

Marked out of 1.00

Which among the following antipsychotic drugs causes the most prolongation of QT-c interval on the ECG?

Select one:

- Clozapine
- Olanzapine
- Haloperidol
- Chlorpromazine
- Thioridazine

Your answer is incorrect.

The effect on QTC prolongation is high for some psychotropic drugs like **thioridazine**, pimozide, sertindole, and any intravenous antipsychotic. Thioridazine prolongs the QTc interval in a dose-dependent manner. The branded product was withdrawn worldwide in 2005 because it caused severe cardiac arrhythmias.

There is a moderate effect on QT-c prolongation for psychotropic drugs such as chlorpromazine, quetiapine, ziprasidone and tricyclic antidepressants.

The correct answer is: Thioridazine

Question 87

Not answered

Marked out of 1.00

Which of the following is not true regarding drugs used to prevent relapse in alcohol dependence?

Select one:

- Acamprosate interacts with disulfiram
- Disulfiram inhibits metabolism of warfarin
- Metronidazole can cause disulfiram-like reactions with alcohol
- Alcohol should be avoided for 1 week following cessation of disulfiram
- Acamprosate acts on NMDA receptors

Your answer is incorrect.

Acamprosate does not interact with disulfiram. It is believed to act through enhancing GABA transmission in the brain. It is licensed to prevent relapse to alcohol use and has been found to have a modest treatment effect.

Disulfiram works by inhibiting the enzyme acetaldehyde dehydrogenase. Alcohol consumption results in increased serum acetaldehyde, causing diaphoresis, palpitations, facial flushing, nausea, vertigo, hypotension, and tachycardia. This aggregation of symptoms is known as the disulfiram-alcohol reaction and discourages alcohol intake.

The correct answer is: Acamprosate interacts with disulfiram

Question 88

Not answered

Marked out of 1.00

Choose one drug that modulates melatonin system directly.

Select one:

- Trazodone
- Zopiclone
- Agomelatine
- Perphenazine
- Galantamine

Your answer is incorrect.

Agomelatine is a new antidepressant with a novel mechanism of action. It acts as an agonist to both the melatonin receptors MT1 and MT2. It is also an antagonist to the 5-HT2C serotonin receptor.

The correct answer is: Agomelatine

Question 89

Not answered

Marked out of 1.00

Which of the following accurately reflects the activity of benzodiazepines?

Select one:

- Reduce GABA transmission
- Block sodium reuptake
- Allosteric modification of GABA receptor
- Substitute for GABA
- Facilitate action potential

Your answer is incorrect.

- Benzodiazepines (BDZs) bind to the gamma sub-unit of the GABA-A receptor.
- Their binding causes an allosteric (structural) modification of the receptor that results in an increase in GABA-A receptor activity. BDZs therefore increase the frequency of channel opening events which leads to an increase in chloride ion conductance and inhibition of the action potential.
- BDZs do not substitute for GABA, which binds at the alpha sub-unit.

(Ref: From CNSforum: Basic and clinical pharmacology, 8th edition. Katzung BG. USA: The McGraw Hill Companies, Inc, 2001:364-381.)

The correct answer is: Allosteric modification of GABA receptor

Question 90

Not answered

Marked out of 1.00

A 62-year-old patient with insomnia and depression develops priapism when trazodone is prescribed. This is possibly mediated via:

Select one:

- Beta-1 agonism
- Alpha-2 antagonism
- Alpha-1 antagonism
- Alpha-2 agonism
- Alpha-1 agonism

Your answer is incorrect.

Alpha-1 blockade causes priapism when using trazodone or risperidone.

The correct answer is: Alpha-1 antagonism

Question 91

Not answered

Marked out of 1.00

Impotence as a side effect of antipsychotics is caused due to which of the following mechanisms?

Select one:

- Alpha-2 adrenergic blockade
- Histaminergic receptor blockade
- Nicotinic receptor blockade
- Alpha-1 adrenergic blockade
- Muscarinic receptor blockade

Your answer is incorrect.

Side effects due to alpha-1 adrenergic blockade include sedation, postural hypotension, impotence and ejaculatory failure.

Alpha-1 adrenergic blockade leads to a moderately high reported incidence of ejaculatory problems such as retrograde ejaculation and rarely priapism. Sexual dysfunction has been reported as a side effect of all antipsychotics and up to 45% of people taking conventional antipsychotics experience sexual dysfunction.

The correct answer is: Alpha-1 adrenergic blockade

Question 92

Not answered

Marked out of 1.00

The mechanism of action of varenicline is:

Select one:

- Alpha-2 beta-4 nicotinic acetylcholine receptor partial agonist
- Alpha-2 beta-2 nicotinic acetylcholine receptor partial agonist
- Alpha-4 beta-2 nicotinic acetylcholine receptor partial antagonist
- Alpha-4 beta-2 nicotinic acetylcholine receptor partial agonist
- Alpha-2 beta-4 nicotinic acetylcholine receptor partial antagonist

Your answer is incorrect.

Varenicline is a partial agonist at the **alpha-4 beta-2 (α4β2) unit of nicotinic acetylcholine receptor**. It assists smoking cessation by relieving nicotine withdrawal symptoms and reducing the rewarding properties of nicotine.

The correct answer is: Alpha-4 beta-2 nicotinic acetylcholine receptor partial agonist

Question 93

Not answered

Marked out of 1.00

Which of the following medications have dual action in prophylaxis of influenza as well as in treatment of Parkinson's disease?

Select one:

- Carbidopa
- Levodopa
- Orphenadrine
- Bromocriptine
- Amantadine

Your answer is incorrect.

Amantadine is an antiviral agent and prevents viral uncoating. It also induces dopamine release from nerve terminals and so is used in Parkinsonism.

The correct answer is: Amantadine

Question 94

Not answered

Marked out of 1.00

Which of the following statement is correct with regard to pharmacokinetics?

Select one:

- Chlorpromazine decreases levels of tricyclics
- Most orally administered antipsychotics undergo first pass metabolism
- Depots achieve peak plasma levels on the third day after injection
- First pass metabolism enables easier entry through blood brain barrier
- Extensive first pass metabolism increases bioavailability

Your answer is incorrect.

- The correct answer is: **Most orally administered antipsychotics undergo first pass metabolism.**
- First pass metabolism will serve to reduce bioavailability.
- It does not influence passage via BBB.
- Chlorpromazine inhibits TCA metabolism thereby increasing TCA levels.

The correct answer is: Most orally administered antipsychotics undergo first pass metabolism

Question 95

Not answered

Marked out of 1.00

What is the half-life of risperidone?

Select one:

- 2 hours
- 5 hours
- 150 hours
- 8 hours
- 24 hours

Your answer is incorrect.

The half-life of risperidone and of its active metabolite is about 24 hours. The apparent half-life of risperidone is 3 hours in extensive metabolisers and 20 hours in poor metabolisers.

The correct answer is: 24 hours

Question 96

Not answered

Marked out of 1.00

Mr. Z is a 43-year-old gentleman who has a 15 year history of schizophrenia and poor adherence with antipsychotic treatments. He has been stabilised on 15mg of olanzapine in the hospital and he has previously done well on flupentixol injections. He has agreed to switch to flupentixol injection once a fortnight. The test dose was administered. What is the most likely amount of time he will need to continue taking the oral olanzapine?

Select one:

- 3-6 months
- 2-3 months
- 2-3 weeks
- 1 month
- 1-2 weeks

Your answer is incorrect.

The depot medication may require 3-6 months to reach a steady plasma level, therefore oral antipsychotic medication should be continued until steady state plasma level is achieved. Attainment of peak plasma levels, therapeutic effect and steady state plasma levels are all delayed with depot injection. The use of adjunctive oral medication to assess depot requirements may be helpful and oral anti psychotic cover is sometimes required for longer periods of time. Moreover, in most clinical trials relapse seems only to occur six months after withdrawing depot therapy; roughly the time required to clear steady state depot drug levels from the blood.

The correct answer is: 3-6 months

Question 97

Not answered

Marked out of 1.00

The dosage of duloxetine used for the treatment of diabetic neuropathy is:

Select one:

- 60-120mg
- 5-10mg
- 10-20mg
- 120-240 mg
- 30-60 mg

Your answer is incorrect.

Duloxetine inhibits the re-uptake of both serotonin and noradrenaline. It is used for the treatment of major depressive disorder (60mg once daily), generalised anxiety disorder (30-120mg) and the treatment of diabetic neuropathy (60-120mg in divided doses).

The correct answer is: 60-120mg

Question 98

Not answered

Marked out of 1.00

Aripiprazole has an elimination half-life of:

Select one:

- 3.5 hours
- 2 hours
- 12 hours
- 24 hours
- 75 hours

Your answer is incorrect.

Aripiprazole is metabolized extensively in humans, with a mean terminal half-life of 60 hours (up to 76 hours) after 14 days of administration. The half-life of its major metabolite, dehydroaripiprazole, is 96 hours.

The correct answer is: 75 hours

Question 99

Not answered

Marked out of 1.00

Nancy is a 58-year-old high school chemistry teacher who has recently been diagnosed with Alzheimer's Dementia. She is keen to undergo genetic testing to know whether her children are at increased risk of the disease. Presenilin-1 is a gene implicated in early onset Alzheimer's Dementia. Mutations in the presenilin-1 gene are located on which one of the following chromosomes?

Select one:

- Chromosome 14
- Chromosome 1
- Chromosome 21
- Chromosome 13
- Chromosome 8

Your answer is incorrect.

Some cases of early onset Alzheimer's dementia (EOAD) are clearly inherited in an autosomal-dominant manner. The beta APP gene on chromosome 21, the PS-1 (presenilin-1) gene on chromosome 14, and the PS-2 (presenilin 2) gene on chromosome 1 have all been characterised as genes in which mutations lead to familial EOAD.

(Ref: Clark RF, Hutton M, Talbot C, Wragg M, Lendon C, Busfield F, Han SW, Perez-Tur J, Adams M, Fuldner R, Roberts G. The role of presenilin 1 in the genetics of Alzheimer's disease. InCold Spring Harbor symposia on quantitative biology 1996 Jan 1 (Vol. 61, pp. 551-558). Cold Spring Harbor Laboratory Press)

The correct answer is: Chromosome 14

Question 100

Not answered

Marked out of 1.00

Mitotic cell division results in the production of which type of cells?

Select one:

- Diploid cells
- Gametes
- Aneuploid cells
- Haploid cells
- Triploid cells

Your answer is incorrect.

Mitosis is cell division to produce two identical daughter cells with a full complement of chromosomes (46 in humans). Meiosis results in gametes or haploid cells.

The correct answer is: Diploid cells

Question 101

Not answered

Marked out of 1.00

Mutations in the progranulin gene are thought to be causative in which of the following disorders?

Select one:

- Frontotemporal dementia
- Alzheimer's disease
- Lewy body dementia
- Parkinson's disease
- CJD

Your answer is incorrect.

Approximately 10% of cases of FTD are inherited in an autosomal-dominant manner. Mutations in the gene of microtubule-associated protein tau (MAPT) was the first identified monogenic cause of FTD. Since then, seven mutations in over ten other genes have been identified, with the progranulin gene (GRN), C9orf72 (Chromosome 9 open reading frame 72), TARDBP (TAR DNA-binding protein), FUS (fused in sarcoma), and UBQLN2 (ubiquilin 2). The progranulin gene is located in the 17q21.32 region and is composed of 13 exons. Progranulin is a neurotrophic growth factor. Progranulin mutation is the most common identified genetic variation associated with FTD.

(Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3281593/)

The correct answer is: Frontotemporal dementia

Question 102

Not answered

Marked out of 1.00

Tom Frost is a 6-year-old boy who has learning disability and a prominent jaw. He frequently demonstrates self-hugging and skin-picking behaviours. Which of the following chromosomal abnormalities is most likely to be the cause?

Select one:

- 17p11.2 microdeletion
- 4p16 deletion
- 5p deletion
- 7q11 microdeletion
- 10q deletion

Your answer is incorrect.

This is likely to be Smith-Magenis syndrome which is a genetic disorder with features including intellectual disability, facial abnormalities, difficulty sleeping, and numerous behavioural problems such as self-hugging and skin-picking. Smith-Magenis syndrome affects an estimated between 1 in 15,000 to 1 in 25,000 individuals. It is a microdeletion syndrome characterised by an abnormality in the short (p) arm of chromosome 17 and is sometimes called the 17p syndrome.

The correct answer is: 17p11.2 microdeletion

Question 103

Not answered

Marked out of 1.00

Mr Richard Atkinson is a 78-year-old man who has been brought to the GP by his family with progressive memory loss, repeated falls and pill rolling tremor. Which of the following is this condition most associated with?

Select one:

- Alpha-synuclein
- Progranulin
- Dysbindin
- Glutamate
- Neuregulin

Your answer is incorrect.

This is most likely to represent Parkinson's disease. This is characterised by cell death in the brain's basal ganglia and the presence of Lewy bodies (alpha-synuclein protein) in many of the remaining neurones. People with parkinson's disease can develop progressive memory loss and eventually develop dementia. The stooped posture can cause balance problems leading to repeated falls in parkinson's disease.

The correct answer is: Alpha-synuclein

Question 104

Not answered

Marked out of 1.00

Genes encoding for which of the following proteins are implicated in ADHD?

Select one:

- DRD4
- DRD2
- MAO-A
- MAOB
- COMT

Your answer is incorrect.

The genome-wide linkage studies conducted to date show divergent findings. Nevertheless, candidate gene studies have produced consistent evidence implicating the genes coding for **DRD4, DRD5, SLC6A3, SNAP-25, and HTR1B** in the aetiology of ADHD. Ref: Faraone SV, Mick E. Molecular genetics of attention deficit hyperactivity disorder.

(Psychiatric Clinics. 2010 Mar 1;33(1):159-80)

The correct answer is: DRD4

Question 105

Not answered

Marked out of 1.00

Which of the following can be considered an example of epigenetic mechanisms operating to influence the risk of mental illnesses?

Select one:

- Early maternal support enhances hippocampal development
- Depression is more prevalent in the aftermath of negative life events
- Risk of psychosis varies with genetic variations in the complement system
- Family history of bipolar disorder increases the risk of both schizophrenia and depression
- Psychosis often emerges from an attenuated symptom state

Your answer is incorrect.

DNA methylation (an epigenetic mechanism) in children raised in institutions is different compared with those raised by their biological parents. In addition, maternal support observed in early childhood is noted to be strongly predictive of hippocampal volume measured at school age.

Ref: Luby JL, Barch DM, Belden A, Gaffrey MS, Tillman R, Babb C, Nishino T, Suzuki H, Botteron KN. Maternal support in early childhood predicts larger hippocampal volumes at school age. Proceedings of the National Academy of Sciences. 2012 Feb 21;109(8):2854-9.

The correct answer is: Early maternal support enhances hippocampal development

Question 106

Not answered

Marked out of 1.00

Mr. Alan Vine is a 45-year-old man who has experienced a major head injury leaving him with a degree of disability. He notes that he can no longer keep his balance on a bike or handle a pen with the dexterity that he used to. Damage to which of following areas of his brain is most likely to account for his symptoms?

Select one:

- Dominant parietal lobe
- Prefrontal cortex
- Dominant temporal lobe
- Non-dominant parietal lobe
- Non-dominant temporal lobe

Your answer is incorrect.

Apraxia or dyspraxia is most likely to be due to a lesion affecting the dominant parietal and/or frontal lobe. It is possible for it to result from damage to the non-dominant side, but this is less common.

The correct answer is: Dominant parietal lobe

Question 107

Not answered

Marked out of 1.00

The reward centre of the human brain is located in:

Select one:

- Nucleus accumbens
- Mesocortical pathway
- Mesolimbic pathway
- Ventral tegmental area
- Prefrontal cortex

Your answer is incorrect.

There are numerous areas associated with the reward system but specific reward centres are thought to be located in the **nucleus accumbens** shell, ventral pallidum, parabrachial nucleus, orbitofrontal cortex and insular cortex.

The correct answer is: Nucleus accumbens

Question 108

Not answered

Marked out of 1.00

Mrs Olivia Nelson is a 74-year-old woman with presents to her GP with sudden onset loss of vision on her left side. On examination, she can see objects and movements on her right side without difficulty with both eyes, but is unable to detect any left-sided stimuli. A lesion in which of the following areas is most likely to be the cause of this symptom?

Select one:

- Optic tract
- Parietal lobe
- Temporal lobe
- Chiasmata
- Optic nerve

Your answer is incorrect.

Homonymous hemianopia is a term that describes the visual field loss on the same side of both eyes. It can result from lesions of the contralateral occipital lobe or optic tract.

- Chiasmal lesions lead to bipolar hemianopia.
- Optic nerve lesions result in monocular blindness.
- Temporal and parietal lobe lesions lead to superior and inferior quadrantanopia respectively.

The correct answer is: Optic tract

Question 109

Not answered

Marked out of 1.00

Working memory is known to be impaired in patients with schizophrenia compared to controls. Which of the following areas is implicated in working memory deficits?

Select one:

- Prefrontal cortex
- Corpus callosum
- Hippocampus
- Amygdala
- Thalamus

Your answer is incorrect.

Unfortunately, these kind of MCQs are commonplace in MRCPsych exams. Working memory (or any other cognitive function, for that matter) cannot be reduced to a single brain region or group of cells. Nevertheless, the function of the **prefrontal cortex**, particularly the **dorsolateral portion**, is thought to be critical for the WM deficits seen among patients with schizophrenia.

Ref: Glahn DC, Ragland JD, Abramoff A, Barrett J, Laird AR, Bearden CE, Velligan DI. Beyond hypofrontality: a quantitative meta-analysis of functional neuroimaging studies of working memory in schizophrenia. Hum Brain Mapp. 2005 May; 25(1):60-9.

The correct answer is: Prefrontal cortex

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